

Object-Oriented Programming with Java

Complete Question Paper Pattern · All Units

■ UNIT 1 — Chapters 1–3 | OOP Fundamentals, Classes & Objects, Inheritance & Polymorphism

Question	Chapter	Topic-wise Pattern
Q1(a)	Ch 1	OOP Features, Java Features, JVM/JDK/JRE, Platform Independence, Garbage Collection, Arrays, Strings (Theory/Scenario)
Q1(b)	Ch 2	Constructors, Constructor Overloading, Copy Constructor, toString(), Objects, Reference Variables (Output Prediction)
Q1(c)	Ch 3	Inheritance, Runtime Polymorphism, Method Overriding, Abstract Class, Dynamic Method Dispatch (Large Program)
Q2(a)	Ch 3	Types of Inheritance, Abstract Class, Method Overriding, super, Dynamic Method Dispatch (Theory)
Q2(b)	Ch 1	Association, Aggregation, Composition, Dependency, UML Class Diagram, Relationship Identification
Q2(c)	Ch 2	Classes & Objects, Constructors, Parameterized Constructor, Method Overloading, Arrays of Objects, Static Members (Programming)
Q3(a)	Ch 2	static, final, this, Access Modifiers, Nested Classes, Garbage Collection (Short Theory)
Q3(b)	Ch 3	Inheritance, super, Overriding, Runtime Polymorphism (Output Prediction)
Q3(c)	Ch 3	Abstract Class, Runtime Polymorphism, Object Class, Inheritance Program / Theory + Justification

■ UNIT 2 — Chapters 4–6 | Packages, Interfaces, Exceptions · Collections · Lambdas & Streams

ESA/ISA Part	Chapter	Topic Type	Most Likely Questions
Q3(a)	Ch 4	Theory	Packages, Interfaces, Exception Handling basics
Q3(b)	Ch 4	Output Prediction	Interface polymorphism, multiple interfaces, exception tracing
Q3(c)	Ch 4	Programming	Packages / Interfaces / Exception Handling (case study)
Q4(a)	Ch 5	Theory	Collections hierarchy, List/Set/Map, collection methods
Q4(b)	Ch 6	Theory	Functional Interface, Lambda Expressions, Streams basics
Q4(c)	Ch 6	Programming	Lambda expression program
Q5(a)	Ch 6	Theory	Stream methods, Reduction, Iterators vs Streams
Q5(b)	Ch 6	Output Prediction	Streams API output (filter, map, sorted, count, etc.)
Q5(c)	Ch 5	Programming	Collections program (LinkedList/ArrayList/HashMap)

■ UNIT 3 — Chapter 7 | Design Patterns

Question	Topic-wise Pattern
Q7(a)	Definition of Design Pattern, Need for Design Patterns, Classification (Creational, Structural, Behavioral), Advantages & Applications (Theory – 4M)
Q7(b)	Explain one design pattern with UML/Class Diagram and Working (Factory Method / Singleton / Strategy / Adapter / Composite / Template Method) (6M)
Q7(c)	Scenario-based implementation – Identify the suitable design pattern, justify the choice, draw the UML/Class Diagram (if required), and write the Java program (10M)

■ Quick Reference — Mark Distribution

Question Type	Format	Marks	Notes
(a) parts	Theory / Short Answer	4M	Definition, classification, short explain
(b) parts	Output Prediction / Diagram	6M	Code tracing, UML diagrams, working
(c) parts	Programming / Case Study	10M	Full programs, scenario-based, justification